Catholic Junior College

JC2 Preliminary Examinations

Higher 2

CANDIDATE NAME	
CLASS	**2T**

PHYSICS 9478/03

Paper 3 Longer Structured Questions **15 September 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class in the boxes at the top of this page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

MARK SCHEME

FOR EXAMINER'S USE	
SECTION A	
Q1	/ 11
Q2	/ 7
Q3	/ 9
Q4	/ 11
Q5	/ 5
Q6	/ 12
SECTION B	
Q7	/ 20
Q8	/ 20
PAPER 3	/ 75
PAPER 2	/ 75
PAPER 1	/ 30
PAPER 4	/ 50
TOTAL (WEIGHTED)	%

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[Turn over

DATA: **Updated for new syllabus 9478**

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left. 8.99 \times 10^{9} \text{ m F}^{-1}\right)$
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} mol^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

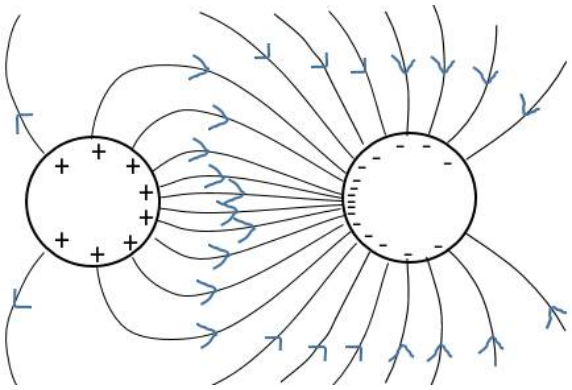
FORMULAE:

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on / by a gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T / K	=	$T / {}^{\circ}C + 273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^2 - x^2\right)}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$

energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0\, e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0\, e^{-\lambda t}$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in this Section in the spaces provided.

1	**(a)**	A positively charged metal sphere A is brought close to a negatively charged metal sphere B as shown in Figure 1.1. The charge on metal sphere B is twice that of the charge on metal sphere A. On Figure 1.1, draw lines to represent the electric field around the spheres.	
			
		Figure 1.1	[2]
	L2	Solution:  Marks scheme: • 1 mark: **Field pattern** (correct shape & variation of density of field lines), **AND**, **Direction** of field lines (arrows point away from A and towards B). • 1 mark: Details: **number** of field lines around B **twice** that around A (**at least 6 field lines around A and 12 field lines around B to sufficiently show trend/pattern**), **AND**, field lines drawn **perpendicular to the surface** of each **metal** sphere. OR • 1 mark: **Field pattern:** correct shape & variation of density of field lines, **number** of field lines around B **twice** that around A (**at least 6 field lines around A and 12 field lines around B to sufficiently show trend/pattern**). • 1 mark: **Direction** of field lines, including field lines drawn **perpendicular to the surface** of each **metal** sphere.	**B1** **B1** [B1] [B1]

Examiner's comments:

- Many candidates did not maintain proper field line density and drew lines which were too sparse or unevenly spaced.
- Some candidates drew field lines entering the interior of the metal spheres rather than ending at the outer boundaries of the spheres.
- Some candidates did not draw directional arrows on several field lines.

(b) Point P is at a distance *x* from the centre of sphere A along the line joining the centres of the two spheres as shown in Figure 1.2. The radius of A and B is 15 mm and the distance between the centres of the spheres is 80 mm.

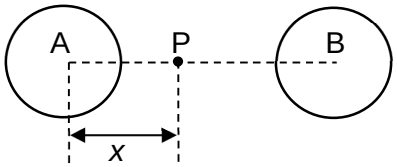


Figure 1.2

The variation with *x* of the electric potential *V* at P is shown in Figure 1.3.

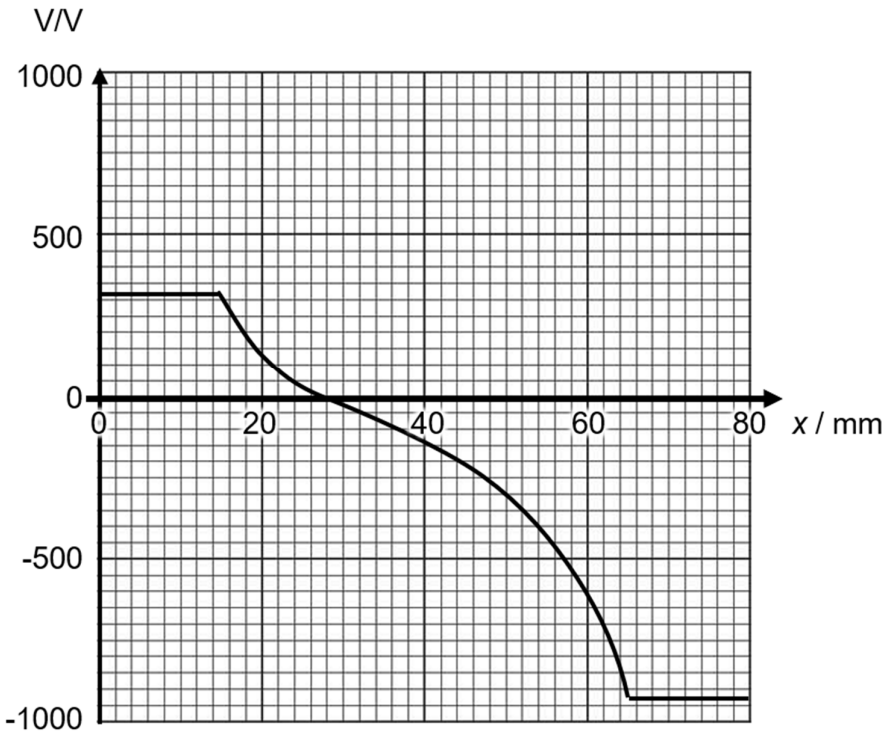


Figure 1.3

(i) Determine the magnitude of the electric field strength at P where *x* = 40 mm.

electric field strength = ... V m^{-1} [2]

		L1	• **Tangent at x = 40 mm** must be **drawn long enough** • **2 coordinates** along the tangent must be **far apart** enough $\|E\| = \left\|\frac{dV}{dx}\right\|$ = magnitude of gradient of V-x graph at x = 40 mm = 350 – (-600) / 80 x 10^{-3} = 1.19 x 10^{4} V m^{-1} • **Accept answers that show correct read-offs, accurate to half-smallest division, of the 2 coordinates.**	M1 A1
		Examiner's comments: • Many candidates were able to obtain full credit for this question. • Some candidates attempted to read values directly from the curve at *x* = 40 mm without drawing tangent lines. • Some candidates did not convert the units for distance *x* from millimeters to meters.			
		(ii)	An electron is released from rest at point P where *x* = 40 mm.		
			1.	Describe and explain the motion of the electron as it travels 20 mm along the line joining the centres of the spheres.	
				.....	
				.....	
				.....	
				.....	
				.....	
				.....	
				.....	[3]
		L2		$a = \frac{F_{net}}{m} = \frac{F_E}{m} = \frac{eE}{m}$ where $E = -\frac{dV}{dx}$ **The electric field strength acts towards the right in the direction of decreasing V, hence the negatively charged electron experiences an electric force acting towards the left.** **Gradient of the V-x graph is proportional to the electric field strength which is proportional to the acceleration.** **When gradient decreases from 40 mm to 28 mm, acceleration decreases.** ➔ **The electron moves to the left towards A, speeds up at a decreasing rate.** **When gradient increases from 28 mm to 20 mm, acceleration increases.** ➔ **The electron continues moving to the left towards A, speeds up at an increasing rate.** *Note that when the electron travels 20 mm from x = 40 mm, it will not yet reach sphere A.*	B1 B1 B1

Examiner's comments:

- Many candidates incorrectly assumed the electron moved towards sphere B rather than towards sphere A.
- A large number of candidates ignored the non-linear curve of the graph and described the motion as having uniform acceleration.
- Some candidates stated that the electron slowed down instead.

2. Determine the speed of the electron when it has travelled 20 mm along the line joining the centres of the spheres.

speed = ……..……m s^{-1} [4]

L2

Moving 20 mm towards A, electron moves from x = 40 mm to x = 20 mm

increase in potential, Δ**V** = **125 – (-150) = 275 V** **C1**

[Also accept V_{40mm} between -125 V to -150 V; V_{20mm} between 125 V to 150 V]

Since the **electron is negatively charge**, it experiences a decrease in electric potential energy when potential increases.

decrease in potential **energy**, Δ**U**
= **|q.ΔV|**
= **$1.60 \times 10^{-19} \times 275 = 4.40 \times 10^{-17}$ J** **M1**

By conservation of energy,
increase in KE = decrease in electric potential energy
final KE – initial KE = ΔU
$\frac{1}{2}$ m $v^2 - 0 = 4.40 \times 10^{-17}$
$\frac{1}{2}$ $(9.11 \times 10^{-31}) v^2 - 0 = 4.40 \times 10^{-17}$ **M1**
$v = 9.83 \times 10^{6}$ m s^{-1} **A1**

Allow partial ECF (award C1 & 1st M1 marks) if candidate in part 1 wrongly deduced that the electron moved to the right from x = 40 mm to 60 mm instead. No mark for 2nd M1 and A1 as it propagates the incorrect energy transformation and so wrong physics.

Examiner's comments:

- Many candidates did not account for the sign change of potential with variation of distance and miscalculated the potential difference.
- Some candidates misread the potential values at *x* = 20 mm and *x* = 40 mm.
- A few candidates substituted incorrect values for the mass or charge of the electron.

[Total: 11]

2 On a computer motherboard where millions of operations can be performed every second, it uses a variety of capacitors for different operations. For example, 100 pF capacitors are used in high-speed areas to filter electrical noises.

(a) Figure 2.1 shows the variation of the charge stored Q on a 100 pF capacitor with potential difference V applied across the capacitor when charging the capacitor.

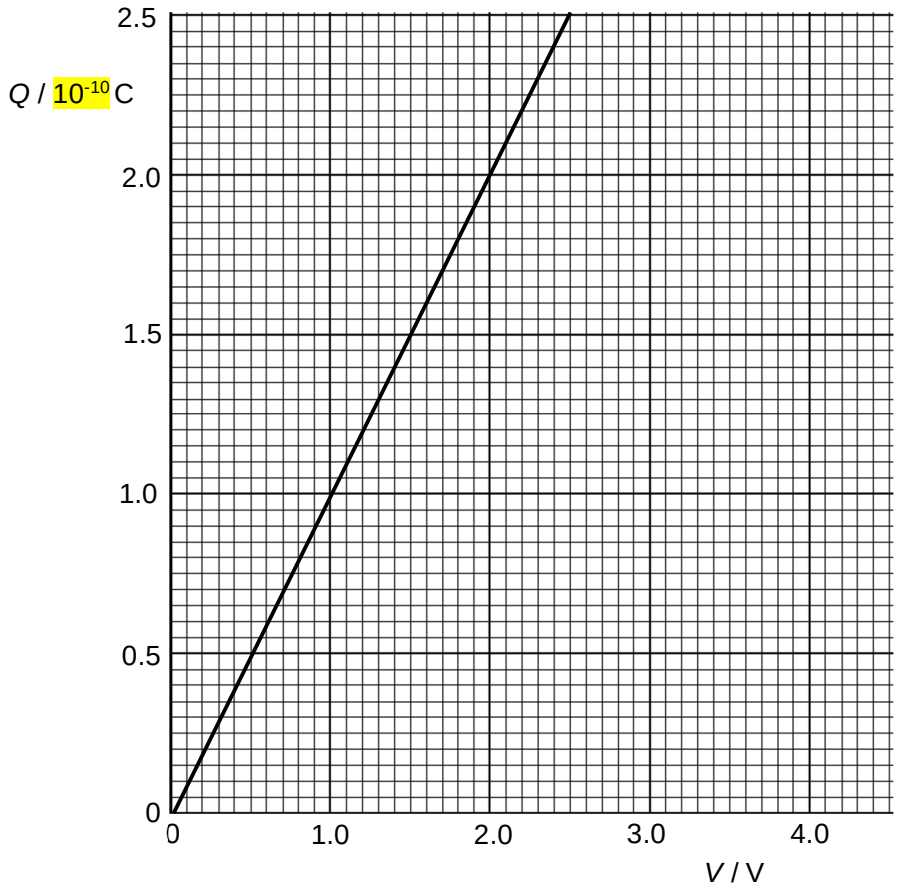


Figure 2.1

(i) Calculate the energy stored in the 100 pF capacitor when the potential difference across it is 2.5 V.

energy stored = ……J [1]

L1 Energy stored = ½ QV
= ½ (2.500 x 10^{-10}) (2.5)
= 3.125 x 10^{-9} J
= 3.13 x 10^{-10} J **A1**

Examiner's comments:
- Many candidates were able to obtain full credit for this question.
- Some candidates overlooked the power-of-ten scale factor (10^{-10} C) on the vertical axis in their calculations.

- Some candidates were unable to convert the value of the capacitor of 100 picofarads into farads.

(ii) While smaller capacitors are not able to store enough energy to meet sudden increase in current demands in circuitries, they are excellent at suppressing electrical noises.

On Figure 2.1, draw a line to show the variation of Q with V for a 25 pF capacitor. [1]

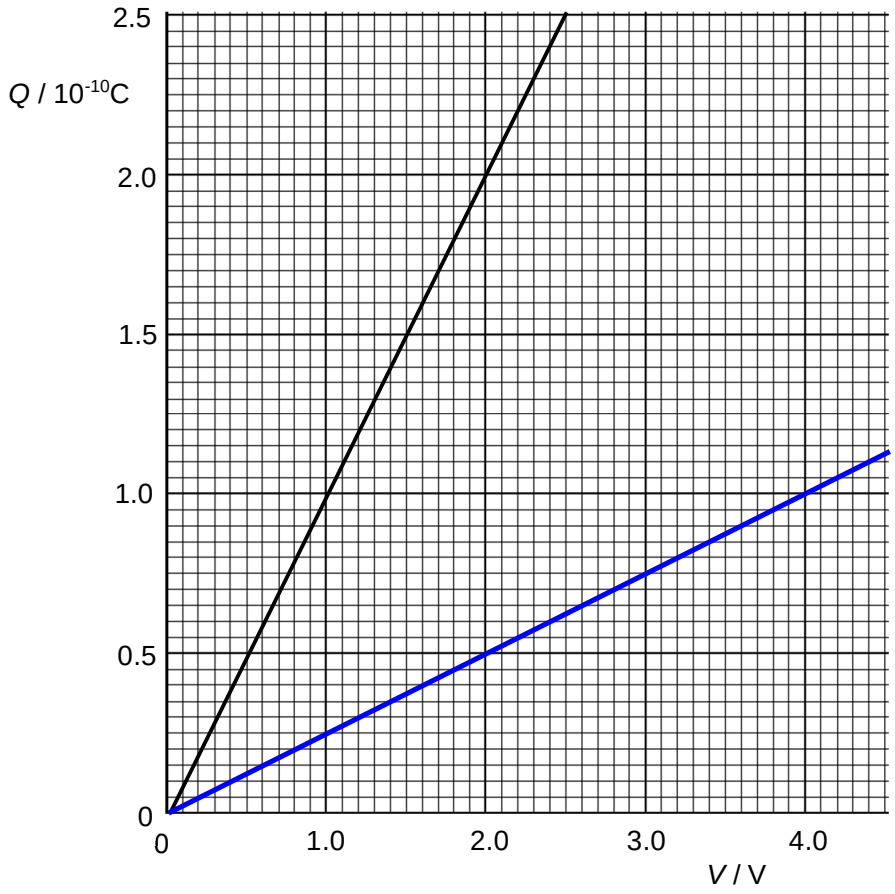
L1

$C = \frac{Q}{V}$

$Q = CV$

∴For a fixed capacitance, graph of Q against V is a straight line passing through (0, 0) with gradient = *C*.

Since 25 pF capacitance is ¼ of 100 pF, **gradient** of the line for 25 pF capacitor should be **one-quarter** that of the line for 100 pF,
e.g. when *V* = 2.0 V, *Q* = 0.5 x 10^{-10} C



B1

Examiner's comments:

- Many candidates drew a curved or exponential graph instead of a straight line.
- Some candidates did not start their line at the origin.
- A few candidates drew a horizontal line showing constant charge independent of the potential difference.

(b) Figure 2.2 shows six identical 1.0 μF capacitors that are connected in parallel. The capacitors are charged and discharged through the same 10 kΩ resistor. This circuit can be used on the computer motherboard to perform the power-up reset command.

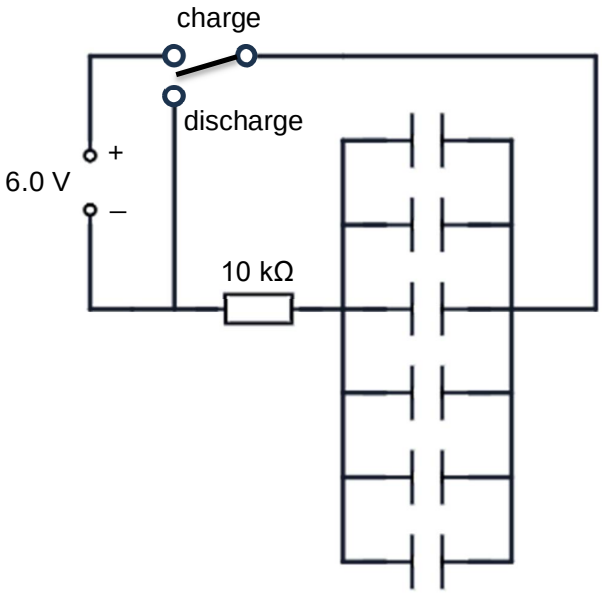


Figure 2.2

The power-up reset command can only be executed when the capacitors are fully charged to 6.0 V, which is equal to the voltage source. When the power-up reset command is executed, the capacitors start to discharge, and the command is completed when the capacitors discharge to 2.0 V or to approximately 33.3% of its maximum voltage V_o. You can take the maximum voltage V_o of the capacitors to be 6.0 V.

(i) Calculate the time taken to complete the power-up reset command.

time taken = ……s [2]

L2

$\tau = RC_e$ = (10 x 10^3) (**6** x 1.0 x 10^{-6}) = 0.060 s

For capacitor that is **dis**charging,

$V_C = V_o e^{-\frac{t}{\tau}}$

$2.0 = 6.0e^{-\frac{t}{0.060}}$ **M1**

$t = 0.065917 = 0.066$ s **A1**

OR

$0.333V_o = V_o e^{-\frac{t}{0.060}}$

$t = 0.065977 = 0.066$ s

Examiner's comments:

- Many candidates applied the wrong formula for the capacitors in parallel and obtained an incorrect value of the total effective capacitance of the circuit.
- Some candidates applied the wrong equation for the discharging capacitor.
- A few candidates expanded the logarithmic expressions incorrectly.

(ii) To protect the delicate circuits on the motherboard, the circuit is designed such that these capacitors can only start to charge again when its potential difference is equal to or less than 0.50 V. Thereafter, the capacitors charge again until they are ready for the next power-up reset command.

Determine the time taken for the capacitors to charge from 0.50 V to full charge. (Assume that a capacitor is fully charged when its potential difference is at 99.3% of its maximum voltage V_o.)

time taken = ……s [3]

L3

Time taken to **charge** from **0.50 V to full charge (~6.0 V)**:

For capacitor **charging**,

$$V_C = V_o\left(1 - e^{-\frac{t}{\tau}}\right)$$

where **time *t* is measured from when V_C = 0**.

∴ Time taken to **charge** from **0.50 V to 6.0 V**
=(Time taken to charge from **0 V to 6.0 V**) – (Time taken to charge from **0 V to 0.50 V**)

Time taken to **charge** from **0 V to 6.0 V**:

$$6.0 = 6.0\left(1 - e^{-\frac{t}{0.060}}\right) \quad \rightarrow t \text{ is undefined}$$

Assume capacitor is fully charged when V_C = 0.993 V_o,

$$0.993V_o = V_o\left(1 - e^{-\frac{t}{0.060}}\right)$$ **M1**

$$t = 0.29771 \text{ s}$$

Time taken to **charge** from **0 V to 0.50 V**:

$$0.5 = 6.0\left(1 - e^{-\frac{t}{0.060}}\right)$$ **M1**

$$t = 0.0052207 \text{ s}$$

Time taken to **charge** from **0.5 V to 6.0 V**
= 0.29771 – 0.0052207
= **0.29249 = 0.29 s** **A1**

Candidates should show understanding that they need to deduct the time taken to charge to 0.5 V.

Examiner's comments:

		• Many candidates applied the wrong equation for charging of a capacitor. • Some candidates assumed the charging rate was linear and attempted to use simple direct proportions between *V* and *t*. • A few candidates substituted the potential difference as 99.3 into the logarithmic expression without dividing it by 100.

[Total: 7]

3 (a) A solenoid shown in Figure 3.1 has length of 10 cm and 300 turns of wire. The current in the solenoid is 2.5 A when it is connected to a power supply.

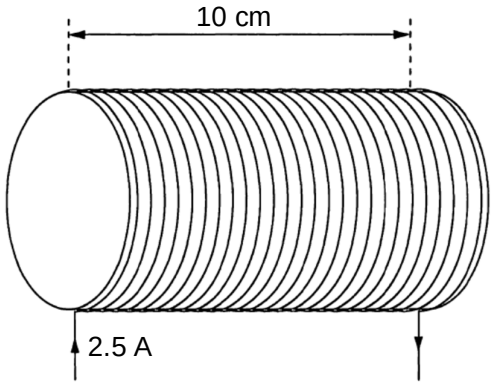


Figure 3.1

(i) Calculate the magnetic flux density inside the solenoid due to current of 2.5 A.

magnetic flux density = ……T [2]

L1 $B = \mu_o nI = (4\pi \times 10^{-7})(300 / 0.10)(2.5)$ **M1**

$= 9.4248 \times 10^{-3}$

$= \mathbf{9.42 \times 10^{-3}\ T}$ **A1**

Examiner's comments:

- Common mistake is wrong calculation of n, e.g. using 300 or using 0.10cm/300 or using 300x0.10cm. Formula is given in formulae sheet, so wrong value of n substituted shows wrong understanding of n and hence M1 mark not awarded. A1 mark not awarded if M1 mark not scored.

(ii) Inside the dashed space provided in Figure 3.2, sketch the magnetic field pattern due to the current in the solenoid when 2.5 A of current is flowing in it as shown. Indicate the polarity of the magnetic field with **N** as the north pole and **S** as the south pole.

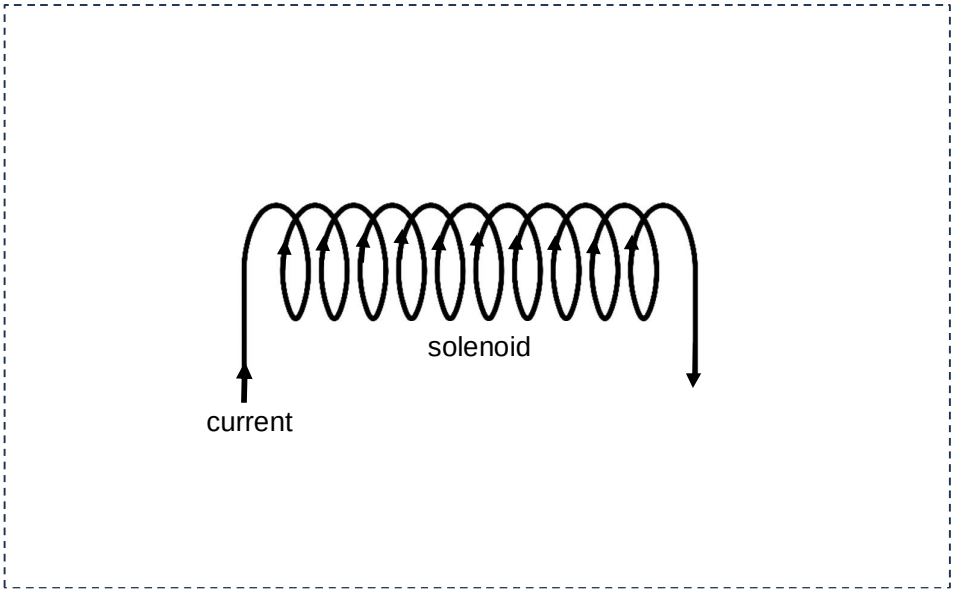
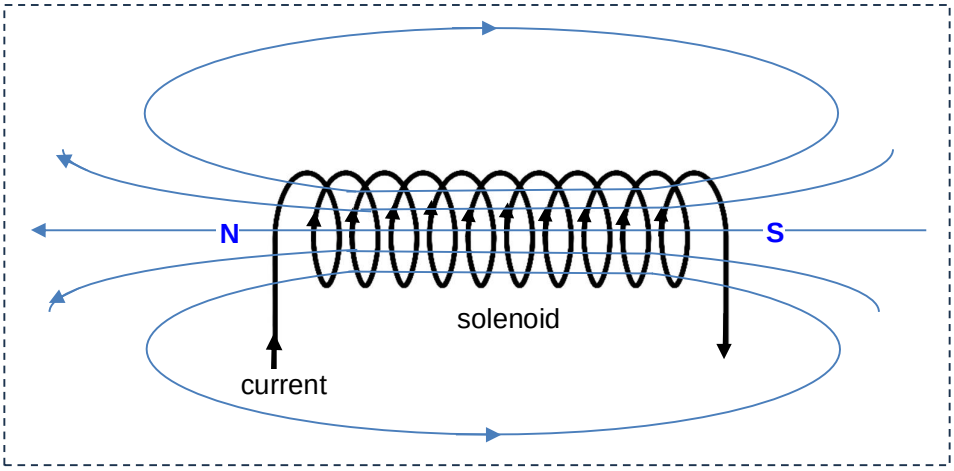


Figure 3.2 [2]

L2 **Solution:**



B1:

- **Correct pattern (correct shape; straight parallel equally spaced fields lines in the centre of the solenoid, diverge towards the ends of the solenoid; separation of field lines closer inside the solenoid compared to outside of it.)**
- **At least 5 lines drawn to show the trend/pattern.**
- **Field lines should not touch one another.**

B1:

- **Correct direction of field lines.**
- **Correct polarity (N and S) indicated.**

Examiner's comments:

- Pattern poorly drawn. Mistakes include not showing a uniform field in the centre within the solenoid (not drawing parallel, straight & equally spaced lines using a ruler), incomplete field lines (showing field lines at the ends but not above & below

the solenoid), too few lines drawn to show the pattern e.g. drawing 2 or 3 field lines, broken lines, lines touching one another.

- Several candidates could not get the direction of lines and/or polarity correct, and so lost the 2nd B1 mark.
- Several candidates did not draw field lines *inside* the solenoid, suggesting no field inside the solenoid.

(b) A pair of concentric coils made of an inner coil Y placed within a solenoid is shown in Figure 3.3. The solenoid has cross-sectional area of 3.0×10^{-2} m^2 and is made of 2500 turns of wire. The solenoid is connected to a variable power supply by the terminals CD.

Coil Y has 500 turns of wire, a cross-sectional area of 7.5×10^{-3} m^2 and resistance of 3.0 Ω. Coil Y is connected to a resistor R of 15 Ω.

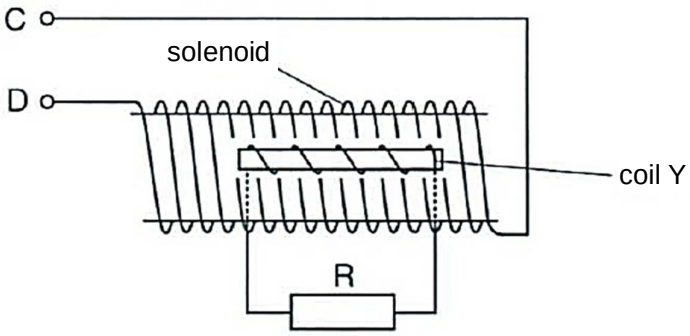


Figure 3.3

The current in the solenoid is then varied causing the magnetic flux density *B* in the solenoid to vary with time *t* as shown in Figure 3.4.

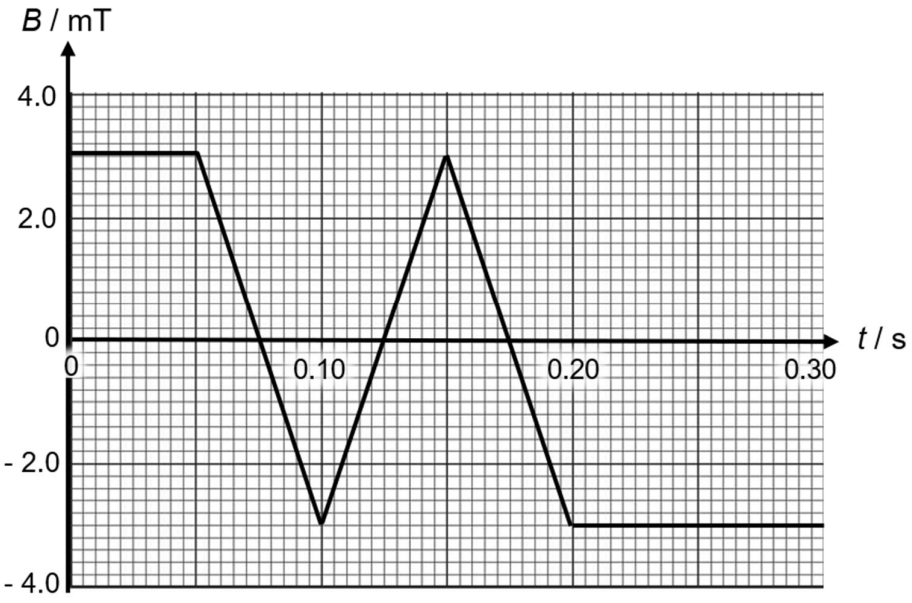


Figure 3.4

(i) Calculate the magnitude of the maximum current in R.

current = ……A [3]

L2

Magnitude of **maximum** Induced EMF

$$E_Y = -\frac{d\Phi_Y}{dt} = -\frac{d\left[N_Y B_S A_Y\right]}{dt} = -N_Y A_Y \frac{dB_S}{dt}$$

$$= -(500)\left(7.5\times10^{-3}\right)\left[\frac{\left(3.0-(-3.0)\right)\times10^{-3}}{0.050}\right]$$ ***(use the maximum gradient)*** **M1**

$$= -0.45 \text{ V}$$

$$|E_Y| = 0.45 \text{ V}$$

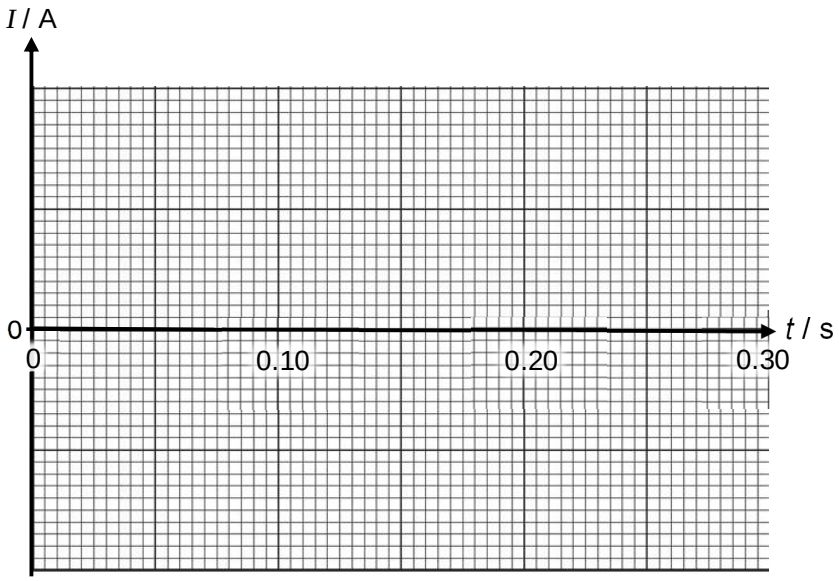
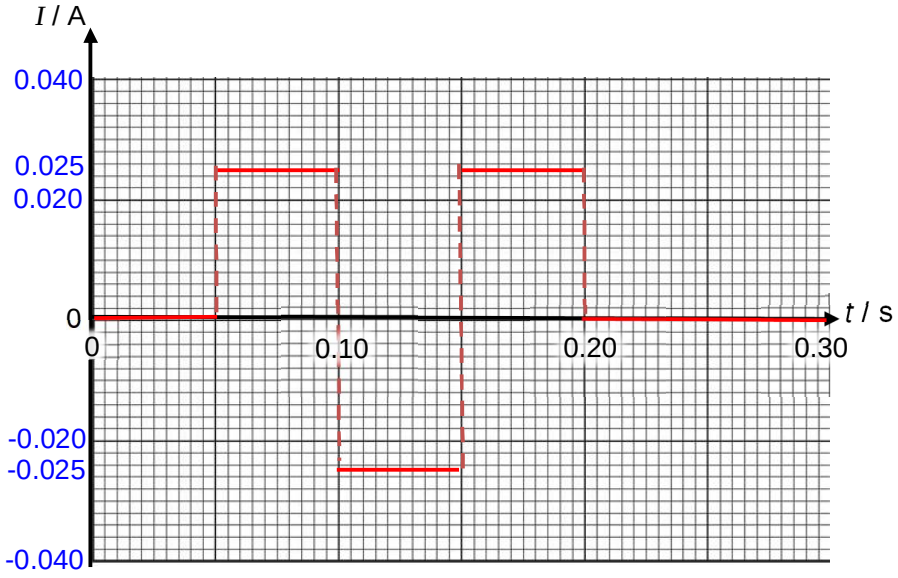
Magnitude of **maximum** current

$$I_{ind} = \frac{E_{ind}}{R_{total}} = \frac{0.45}{3.0+15}$$ **M1**

$$= 0.025 \text{ A}$$ **A1**

Examiner's comments:

- Many candidates could not determine the induced e.m.f. correctly, e.g. using B = μ_onI instead of Faraday's law; confusing the number of turns and cross-sectional areas for the outer solenoid and inner coil Y.
- Many candidates did not include the resistance of coil Y (3.0 Ω) when determining the induced current. Some candidates thought 3.0 Ω and 15 Ω resistances are in parallel rather than series.
- Some candidates committed Power of Ten (P.o.T.) error, e.g. not converting mT to T – deducted under A1 mark.

		(ii)	On Figure 3.5, use your answer in **(b)(i)** to sketch a graph to show the variation with time of the current I in resistor R for time t = 0 to t = 0.30 s.
			
		Figure 3.5	[2]
	L2	**Solution:** 	
		B1: • **Where I = 0 from t = 0 to t = 0.05 s, and, t = 2.0 s to t = 3.0s.** **B1:** • **Maximum value of I from (b)(ii) is plotted accurately to ½ the smallest division. [Allow ECF.]** • **Where $\frac{dB}{dt}$ is constant and positive, I is constant and negative, and** • **Where $\frac{dB}{dt}$ is constant and negative, I is constant and positive.** • **Good scale for y-axis: 1:1, 1:2, 1:4, 1:5 and graph covers at least half the graph grid vertically.**	

		Examiner's comments: • Most candidates scored the 1st B1 mark. • Many candidates did not get the 2nd B1 mark due to awkward scale and/or wrong variation or not accounting for Lenz's law for the direction of the induced current.

[Total: 9]

4 **(a)** Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Figure 4.1. The effective spring constant of the combined springs is 11 N m^{-1}.

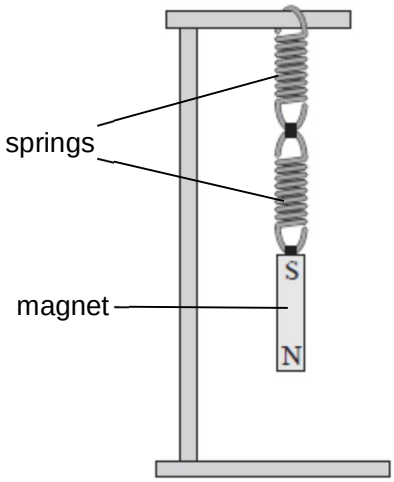


Figure 4.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

(i) By considering the forces on the magnet at its equilibrium position and also at a small displacement x downward from its equilibrium position, show that, when released, the magnet's acceleration a is related to its displacement x by the equation:

$$a = -92x$$

Explain your working.

[3]

		L2	**Take direction downward as positive.** **At equilibrium position,** **Upward spring force = Downward weight** **So, Net force = 0.** **At displacement x (downward) below the equilibrium position, spring force increases while weight is unchanged, so** **Upward spring force > Downward weight.** **Net force = -(increase in spring force) = -kx** (negative direction since Net force acts upwards) **By Newton's 2nd law of motion,** **Net force = ma** **-kx = ma** $\mathbf{a = -\frac{k}{m}x}$ $\mathbf{a = -\frac{11}{120 \times 10^{-3}}x}$ a = – 91.667 x a = – 92 x (shown) OR **At Equilibrium position, no net force.** **Upward Spring force = Downward Weight** **$k x_o = mg$ where x_o: spring extension at Equilibrium —(1)** **Take direction downward as positive.** **At displacement x (downward) below the equilibrium position, spring extension is now ($x_o + x$), and net force is non-zero.** **Net force = - (New Spring force) + Weight** **= - k(x_o + x) + mg —--(2)** **Sub (1) into (2):** **Net force = - k(x_o + x) + kx_o = -kx** **By Newton's 2nd law,** Net force = ma **-kx = ma** $\mathbf{a = -\frac{11}{120 \times 10^{-3}}x}$ a = – 91.667 x a = – 92 x (shown) ***Marks scheme:*** ***1 mark: Apply Hooke's law for Spring force. (Must show correct relationship between spring force and extension in context, i.e. do not accept F_{spring} = kx, since x is the INCREASE in extension and NOT the total extension. Note that there is extension at equilibrium too.)*** ***1 mark: Sign convention (show clearly how "-" sign appears).*** ***1 mark: Apply Newton's 2nd law.*** ***Deduct mark if no/inadequate explanation shown to support working for applications of the laws. E.g. Simply writing "F = ma" and "F = -kx" and***	**M1** **M1** **M1**

			combing them does NOT show understanding of what "F" means in context. F in F = ma represents Resultant force, while F in F = -kx represents the INCREASE in Spring Force when displaced by x downward from equilibrium. In fact, total spring force is not kx, since x is not the total extension. Hence candidates must explain working and symbols used clearly!	
		Examiner's comments: • **Many candidates did not explain their working clearly,** e.g. using undefined short forms like F, without stating what they intend F to represent in their proof; did not show/explain clearly that the **Net** force at displacement x downwards is the **Increase in** spring force. • Many did not explain how the "-" sign appears. Answers stating that in SHM acceleration/net force acts opposite direction to x are not accepted since this question requires a **proof** of why acceleration & x are opposite in direction, starting from fundamentals of vector directions and Newton's 2nd law. • **Several candidates stated "spring force = kx" which is incorrect, since x is not the total extension but the increase in extension.** • **Several candidates stated "net force = spring force" which is incorrect, as weight acts too!** • **Some candidates stated that at equilibrium position, the only force acting is the weight, unaware that the spring is extended even at equilibrium position.** • Some candidates used $\omega^2 = \frac{k}{m}$ to get 92. No credit given since candidates are asked to prove the equation, hence derivation from fundamental principles of Newton's 2nd law & Hooke's law is expected.		
		(ii)	Using the expression in **(a)(i)**, calculate the angular frequency of oscillations of the magnet.	
			angular frequency = ……rad s^{-1}	[2]
		L1	a = – 92 x **Comparing with the SHM equation, $a = -\omega^2 x$,** **$\omega^2 = 92$** **M1** $\omega = \sqrt{92}$ **ω** = 9.5917 = **9.6 rad s^{-1} (or 9.59 rad s^{-1})** **A1** *Do NOT accept the following since candidates are expected to derive from expression in (a)(i).* ω = $\sqrt{\frac{k}{m}}$	
		Examiner's comments: • Many candidates did not show clearly why they equate ω^2 to 92. • Several candidates determined f (frequency) instead of ω (angular frequency).		
		(iii)	The amplitude of the oscillations of the magnet is 3.2 cm. The magnet first passes through the equilibrium position at time *t* = 0. Calculate the magnitude of the displacement of the magnet from its equilibrium position when *t* = 0.40 s.	

displacement = ……cm [2]

L2 Take downward as positive direction, and, since x = 0 at t = 0,
Use $x = -x_0 \sin(\omega t)$
x = – (3.2 cm) sin (9.5917 × 0.40) ***Use Radians mode! Allow EFC from (a)(ii).*** **M1**
x = + 2.0494 cm

Magnitude of displacement is **2.0 cm (or 2.05 cm)** **A1**

Examiner's comments:

- Several candidates did not use the Radians mode in their calculator.
- Many candidates used the correct equation relating x and t. However some used cosine instead of sine. Others assumed a proportional relationship between x and t forgetting it is SHM!

(b) Two identical systems shown in Figure 4.1 are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting wires as shown in Figure 4.2.

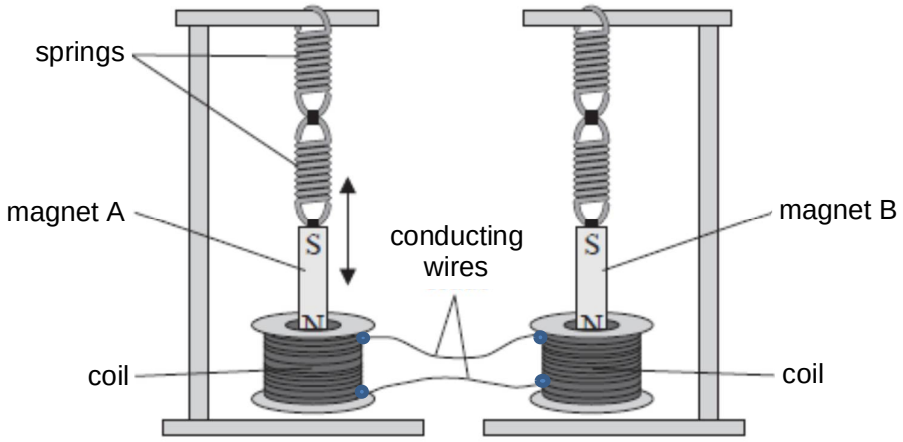


Figure 4.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil as it oscillates. As this motion continues, magnet B starts to oscillate. The amplitude of the oscillations of magnet B increases over time.

Explain why magnet B starts to oscillate and why the amplitude of oscillations increases.

……

……

……

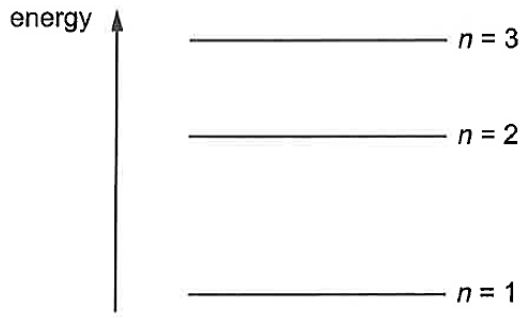
……

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		……...	
		……...	
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		……...	
		……...	
		……...	
		……	[4]
	L3	**As the North-pole of magnet A moves into and out of the left coil, the coil experiences alternating increase and decrease in magnetic flux density and hence changes in magnetic flux linkage Φ.** **By Faraday's law, this induces an electromotive force (e.m.f.) in the coil** (of magnitude proportional to the rate of change of Φ). **By Lenz's law, the direction of the induced e.m.f. in the left coil (below magnet A) reverses periodically.** **This causes an alternating current to flow in both coils** (since the coils are connected to each other in a closed circuit). **The alternating current produces a magnetic field in the right coil (below magnet B) that reverses direction periodically, and thus attracts and repels magnet B periodically,** driving it into forced oscillations. **Since both spring-magnet systems have the same natural frequency of oscillations, resonance occurs** and thus magnet B oscillates with increasing amplitude.	**B1** **B1** **B1** **B1**
		Examiner's comments: • Several candidates did not understand Faraday's law & Lenz's law thoroughly. E.g. saying a changing flux density induces current, without linking to changes in flux linkage or talking about emf being induced before current can flow; stating other quantities being opposed rather than the change in flux linkage of coil below A that is opposed; mixing up Faraday's & Lenz's law altogether! • Several candidates applied Lenz's law to the attraction & repulsion of magnet B, which alone does not explain the increasing amplitude of B's oscillations. Note that B was not oscillating initially, as such saying that "the top of the coil will act as a north pole to repel the magnet B as it enters the coil, and as a south pole to attract the magnet B as it leaves the coil" makes no sense, since B was not initially oscillating! Note also that Lenz's law would result in decreasing amplitude of A's oscillations over time to oppose the changes that induced the e.m.f., however the goal of the question is to explain why B's oscillations increase in amplitude. Candidates should note that, by Lenz's law, it is the motion of A (not B) that will be opposed. • Several candidates used Fleming's Left Hand Rule to explain the force on magnet B, which is incorrect. FLHR is to determine the direction of magnetic force on a current-carrying conductor or moving charge, not on a magnet. • Many candidates obtained one or more of the first three B1 marks. • A small handful of candidates obtained the last B1 mark.	

[Total: 11]

5	(a)	Explain how line spectra, together with the concept of photon, provide evidence for discrete energy levels in atoms.	
		…….....	
		…….....	
		……	[2]
	L1	**For each emitted photon, the energy is equal to the <u>difference</u> between the energy levels in the atom across which de-excitation of the atom occurred.**	**B1**
		The line spectra only show photons of certain allowed energies and this proves that the energies in an atom can only possess discrete and not continuous values.	**B1**
		Examiner's comments: • Many candidates confused line spectra with photoelectric effect and wrongly discussed threshold frequency and one-to-one interaction. • Students must avoid using inaccurate terms such as energy gaps, particles, the discrete wavelength is the energy gap etc.	
	(b)	A discharge lamp contains excited atoms which emit electromagnetic radiation at several wavelengths. Figure 5.1 shows the three lowest electron energy levels of one of these atoms, labeled n = 1, 2 and 3.  **Figure 5.1** Electron transitions between these three energy levels can produce three different wavelengths of radiation. The transition from n = 2 to n = 1 produces radiation of wavelength 440 nm. Radiation of wavelength 590 nm is also emitted.	
		(i) Show that wavelength of the radiation emitted from the third transition is 250 nm.	
			[2]

		L2	$E_{photon} = \Delta E$ $\frac{hc}{\lambda} = \Delta E$ **590 nm photon** must be emitted from the **smallest** energy transition ***n* = 3 to *n* = 2.**	B1
			The **third transition** must therefore be from ***n* = 3 to *n* = 1**. Let the wavelength of the photon emitted from the third transition be λ_3.	
			$\Delta E_{3\to1} = \Delta E_{3\to2} + \Delta E_{2\to1}$ $\frac{hc}{\lambda_3} = \frac{hc}{590\text{ nm}} + \frac{hc}{440\text{ nm}}$	M1
			$\frac{1}{\lambda_3} = \frac{1}{590\text{ nm}} + \frac{1}{440\text{ nm}}$ $\lambda_3 = 252\text{ nm} \approx 250\text{ nm}$	A0
			Notes: $E_{photon} = \Delta E$ $\frac{hc}{\lambda} = \Delta E$ *The longer the wavelength λ, the smaller the electron energy transition in ΔE the atom.* *Only 3 energy transitions are possible among these 3 energy levels. Thus, since the 440 nm photon is emitted from the transition n = 2 to n = 1, the 590 nm photon must be emitted from the transition n = 3 to n = 2. The third energy transition must therefore be from n = 3 to n = 1.* *While the sum of energies of the photons from the first two transitions should be equal to the energy of the photon from the third energy transition, the wavelengths do not add up because* $\lambda \propto \frac{1}{\Delta E}$ *.*	
		Examiner's comments: • Majority of candidates are able to do this question. • Some candidates associate with the energy of the particle trapped in the infinite potential well.		
		(ii)	State the region of the electromagnetic spectrum in which the third wavelength occurs.	
			……	[1]
		L2	**Ultraviolet** ***Do not accept UV or wrong spelling.***	B1
		Examiner's comments: • Many students are unable to recall the range of wavelengths in the EM spectrum.		

[Total: 5]

6 (a) The isotope beryllium-7 $\left({}^{7}_{4}\text{Be}\right)$ is radioactive with a half-life of 53 days. A beryllium-7 nucleus decays by the emission of a γ-ray photon of energy 0.48 MeV.

(i) Calculate the change in mass of a beryllium-7 nucleus when it undergoes γ-decay.

change in mass = ……kg [2]

L1 Recap:
- Mass-energy of the nucleus is converted into electromagnetic radiation energy (photon).
- Thus, mass of nucleus decreases after it emits a photon.
- Nucleus is heavier in the excited state.

Using Einstein's mass-energy equivalence $E = mc^2$:
(Energy of photon emitted) = (change in mass) c^2
$0.48 \times 10^{6} \times 1.60 \times 10^{-19}$ = (change in mass) $(3.00 \times 10^{8})^2$ **M1**
(change in mass) = 8.5333×10^{-31} = **8.5×10^{-31} kg (or 8.53×10^{-31} kg)** **A1**

Marks scheme:
Wrong conversion of MeV to J is penalized under A1. Award M1 for correct application of (Energy of photon emitted) = (change in mass) c^2

Examiner's comments:
- Almost all students are able to do this question.
- Minority of students did not pay attention to the prefix Mega and the unit electronvolt.

(ii) Determine the probability of decay of a beryllium-7 nucleus in a time of 1.0 day.

probability = …… [2]

L2 Recap: The **decay constant** is defined as the **probability** of a radioactive nucleus decay in the **next unit time interval**.

probability of decay in one day
= the decay constant λ with a unit of day^{-1} = $\dfrac{\ln 2}{53 \text{ days}}$ **M1**
= 0.013078 = **0.013** day^{-1} **(or 0.0131** day^{-1}**)** **A1**

Do not penalise if unit is not stated in the final answer.

Additional note (not in A-level syllabus):
more correctly,
Probability of decay in 1-day = $1 - e^{-\lambda(1)} \approx \lambda$
Probability of decay in x-days = $1 - e^{-\lambda(x)} \approx \lambda$
(this expression $P \approx \lambda$ is true as long as λ and x are small)

Another way to understand probability of decay per unit time and half-life, after 53 days, $e^{-(0.0131)(53)} = 0.50 = 50\%$. 50% of the radioactive sample remain undecayed, corresponding to the half-life of 53 days.

Examiner's comments:

- Some candidates did not realise that since the half-life is 53 days, by finding the decay constant, one is finding the probability of decay per unit time (day), which is the rate of decay per day.

(iii) Initially, the total mass of undecayed beryllium-7 in a radioactive sample is 5.7×10^{-12} kg.

Calculate the number of undecayed beryllium-7 nuclei in the sample after 120 days.

number = …… [3]

L2

number of undecayed nuclei
$N = N_o e^{-\lambda t}$

where N_o = number of undecayed nuclei at time t = 0
= initial mass of sample / mass of a nucleus

$$= \frac{5.7 \times 10^{-12} \text{ kg}}{7 \text{ u}}$$

$$= \frac{5.7 \times 10^{-12} \text{ kg}}{7 \times 1.66 \times 10^{-27} \text{ kg}}$$

M1

number of undecayed nuclei
$N = N_o e^{-\lambda t}$

$$= \frac{5.7 \times 10^{-12} \text{ kg}}{7 \times 1.66 \times 10^{-27} \text{ kg}} e^{(-0.013078 \times 120)}$$

M1

$= 1.0212 \times 10^{14} =$ **1.0×10^{14} (or = 1.02×10^{14})**

A1

Marks scheme:
Wrong / Lack of conversion of u to kg is penalized under A1.
Award 1st M1 for evidence of knowledge that 7u is the nuclear mass.
Award 2nd M1 for correct application of $N = N_o e^{-\lambda t}$.

Examiner's comments:

- If students want to find the number of moles, students should note that they are dividing the total mass by 7g and not 7u. In that case, the total mass should also be converted to grams.
- Awareness of units is poor. Some students apply the exponential equation / decay to the mass of the sample and state that as the final answer when the question is clearly asking for number of nuclei.

(iv) Explain why, although the beryllium-7 is radioactive, the number of beryllium-7 nuclei in the sample does not change.

……

……

……

…… [2]

L2 A **γ-ray photon is a quantum of** (electromagnetic radiation) **energy** (and has no proton and neutron). **B1**

When a Be-7 nucleus emits a γ-ray photon, its proton number and neutron number are unchanged. The type of nuclide thus remains unchanged. (OR, The type of nucleus depends on the proton number and the neutron number in the nucleus. Thus the Be-7 nucleus remains as Be-7 nucleus after the emission of a γ-ray photon.) **B1**

Examiner's comments:

- Most students are able to identify that the nucleus emits a γ-ray photon. Students need to explain what is γ-ray photon i.e. no proton and no neutron, which is essential for this question.
- Many students mixed up nuclei with nucleons.

(b) The beryllium-7 nucleus in **(a)** may be considered to be stationary before the emission of the γ-ray photon.

Determine the recoil speed of a beryllium-7 nucleus due to the emission of a γ-ray photon.

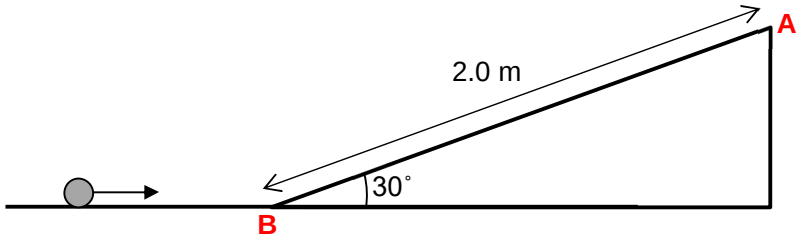
speed = ……m s^{-1} [3]

	L2	From de Broglie's relation, $p = \frac{h}{\lambda}$ Energy of a photon, $\mathbf{E = hf = \frac{hc}{\lambda} = \left(\frac{h}{\lambda}\right)c = pc}$ ∴ Momentum of a photon $p = \frac{E}{c}$ $= \frac{\mathbf{0.48 \times 10^{6} \times 1.60 \times 10^{-19}}}{\mathbf{3.00 \times 10^{8}}}$ $= 2.56 \times 10^{-22}$ kg m s^{-1} Initial total momentum = 0 since the Be-7 nucleus is at rest initially Final total momentum = (momentum of Be-7) + (momentum of photon) **Take direction of photon's momentum to be positive direction.** Let v be the recoil speed of Be-7. By conservation of total linear momentum, Final total momentum = Initial total momentum (momentum of Be-7) + (momentum of photon) = 0 **(-7u.v) + (+2.56 x 10^{-22}) = 0** **7 (1.66 x 10^{-27}) v = 2.56 x 10^{-22}** **v = 22000 m s^{-1}** ***Marks scheme:*** ***Wrong conversion of MeV to J is penalized under A1. [Allow ECF from (a)(i) for wrong conversion of MeV to J.]*** ***Wrong / Lack of conversion of u to kg is penalized under A1.*** ***Award 1st M1 mark for correct values substituted albeit wrong units conversion.*** ***Award 2nd M1 mark for correct application of conservation of momentum with sign convention considered.***	**M1** **M1** **A1**
		Examiner's comments: • This is a challenging question for many students. Many students did not realise energy is not conserved. • Strong students are able to identify conservation of momentum as the key point to the solution. • However, a mark is taken away if the idea of COM is not shown clearly or if the opposite directions of the velocities are not shown in the working.	

[Total: 12]

Section B

Answer **one** question in this Section in the spaces provided.

7	**(a)**	State *Newton's second law of motion*.	
		……...	
		……...	
		……	[2]
	L1	**The rate of change of momentum of a body is (directly) proportional to the resultant force acting on the body**	**B1**
		and is in the same direction as the resultant force.	**B1**
		Examiner's comments: • Poorly done by majority of candidates. Please spend some time memorizing formula. • Correct answers need to relate to rate of change of momentum to RESULTANT force, and not simply force. • Not acceptable to state 'Rate of change of momentum over time or relative to time". Also not acceptable to say change of momentum over time (following definition in syllabus). • A number of candidates miss out the second mark on direction.	
	(b)	A rollerblader approaches and glides up a smooth ramp of length 2.0 m along AB, inclined at an angle of 30˚ to the horizontal. The side view of this ramp is shown in Figure 7.1.  **Figure 7.1** Figure 7.2 shows the ramp when viewed **perpendicular** to the ramp's inclined surface. The rollerblader glides up the ramp with initial speed u, at an angle of $\theta = 60^{\circ}$ to the base of the ramp as shown in Figure 7.2. He stops exerting energy the moment he starts moving up the ramp.	

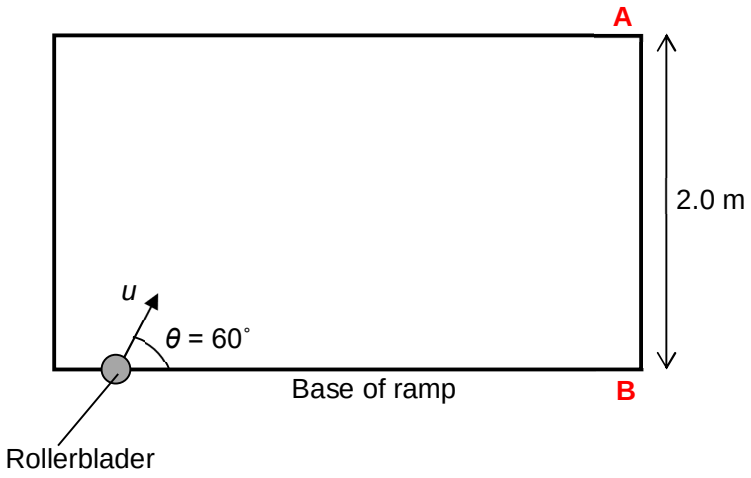
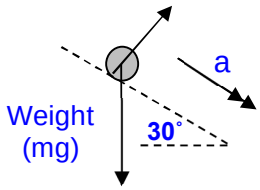


Figure 7.2

(i) Determine the magnitude of the acceleration of the rollerblader while moving up the ramp.

acceleration = ……m s^{-2} [2]

L1



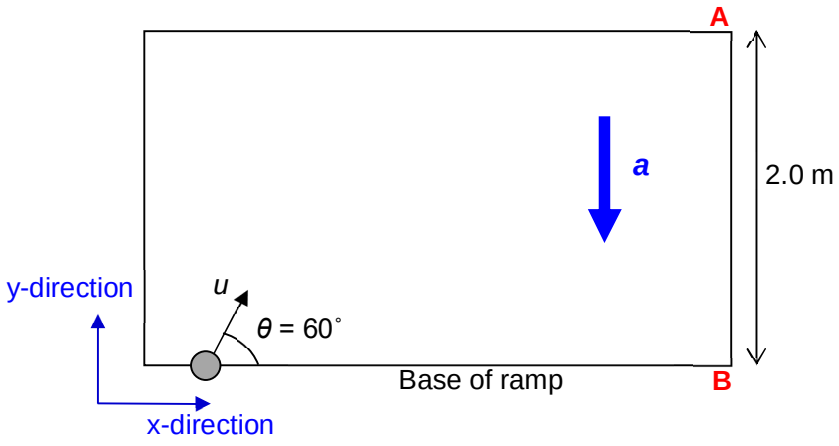
By Newton's 2nd law of motion,
mg sin30˚ = ma **M1**

a = g sin30˚ = (9.81) sin30˚
= 4.905 = **4.9 m s^{-2} (or 4.91 m s^{-2})** **A1**

Examiner's comments:
- Generally well done.

(ii) On Figure 7.2, draw an arrow to show the direction of the acceleration of the rollerblader. [1]

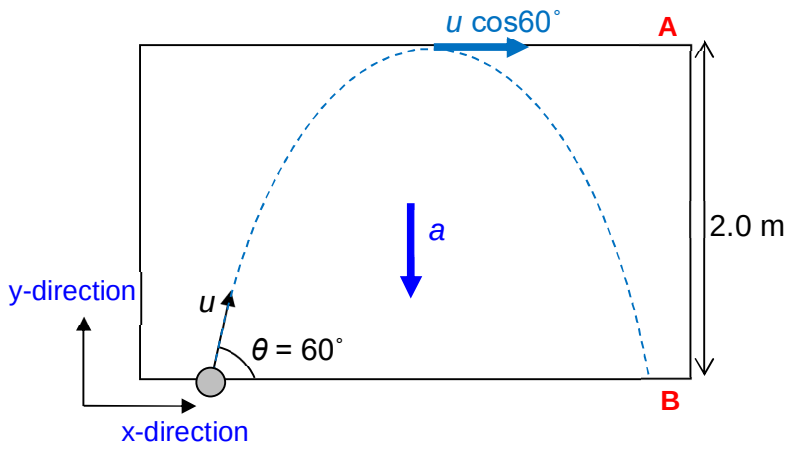
L1 Solution:



B1

		Examiner's comments: • Most candidates can indicate correct direction. A number of candidates drew on the object.		
		(iii)	Explain why the rollerblader does not move up the ramp along a straight path but along a parabolic path.	
			…….....	
			…….....	
			…….....	
			…….....	
			…….....	
			…….....	[3]
		L2	**The acceleration is constant in magnitude and direction throughout the motion,** **and acts non-parallel to the initial velocity.** **This causes the rollerblader to travel with uniform velocity in the direction parallel to the base of the ramp / in the x-direction, and,** **uniform acceleration in the perpendicular direction to base of ramp / in the y-direction.** Hence the rollerblader traces a parabolic path. *(similar to projectile motion in a gravitational field when a particle is projected at an angle to the acceleration due to gravity)*	**B1** **B1** **B1**
		Examiner's comments: • Most candidates are able to answer in terms of velocity or acceleration in two components. • Some candidates give vague answers attributing acceleration due to gravity (when actually it is the component along the slope). • Very few include discussion on acceleration being non-parallel to direction of velocity. Hence most obtain the last two B1 marks.		
		(iv)	By using the principle of conservation of energy, determine the minimum speed *u* for the rollerblader to just make it to the top of the ramp.	
			minimum speed = ……..……m s^{-1}	[3]

L3 Solution:



Loss in Kinetic Energy (KE) = Gain in Gravitational Potential Energy (GPE)
$KE_i - KE_f = mgh$
$\frac{1}{2} mu^2 - \frac{1}{2} m(u\cos 60^\circ)^2 = mg(2.0 \sin 30^\circ)$
$u^2 - (u\cos 60^\circ)^2 = 2g(2.0 \sin 30^\circ)$
u = 5.1147 = **5.1 m s^{-1} (or 5.11 m s^{-1})**

- ***M1 for correct expression for Change in KE_f.*** **M1**
- ***M1 for correct expression for Change in GPE.*** **M1**
- ***A1 for correct final answer.*** **A1**

Examiner's comments:

- Majority of candidates fail in identifying the expression for change in KE. Change in KE is only in the y direction. Hence most candidates took the answer in their working the initial velocity when it is actually the initially velocity in the y direction.
- Majority of candidates are able to obtain the mark for GPE.

(v) Using the speed in **(b)(iv)**, determine the time taken for the rollerblader to return to the base of the ramp.

time taken = ……s [2]

L2 return to the base → **$s_y = 0$**

Take downslope as negative direction.

Use $s_y = u_y t + \frac{1}{2} a_y t^2$
$0 = (+5.1147\sin 60^\circ)\, t + \frac{1}{2} (-4.905)\, t^2$ **M1**
t = 1.8061 = **1.8 s (or 1.81 s)** **A1**

Examiner's comments:

- Poorly done as a lot of candidates did not solve the question by considering the y-direction only i.e used the initial velocity, u , obtained in b(iv) without resolving it to u_y.
- A number also made the mistake of not assigning signs to the opposing direction of a_y and u_y.
- Full ecf given if candidates correctly resolve direction of u, a and included correct sign in working with correct calculation.

(c) The ramp is extended to allow the rollerblader to move through a partial vertical circular motion with radius 3.0 m. Figure 7.3 shows the side view of the entire ramp. The rollerblader is upside-down at the top of the ramp.

The rollerblader now approaches the base of the ramp straight on perpendicularly.

You may regard the rollerblader as a point mass throughout this motion.

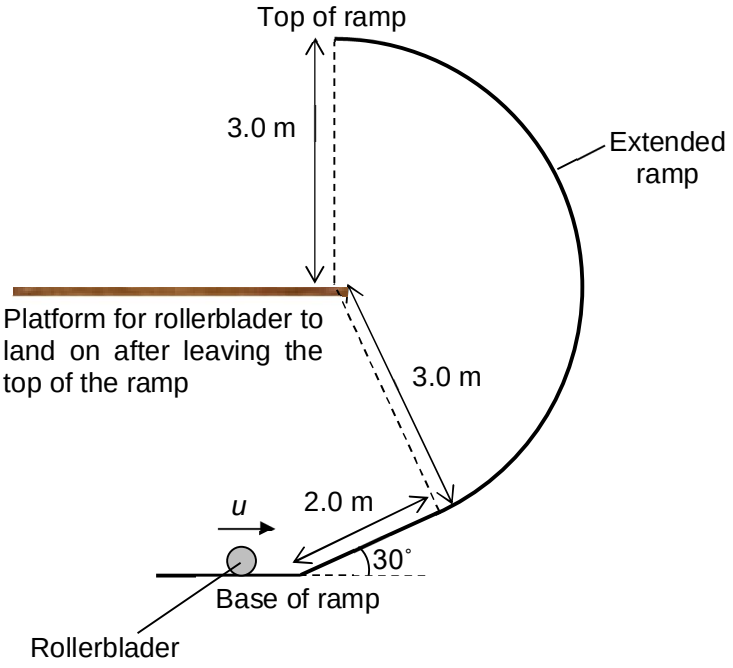
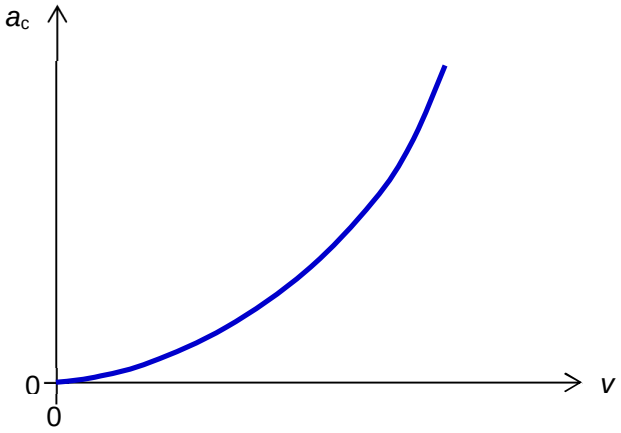


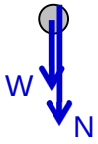
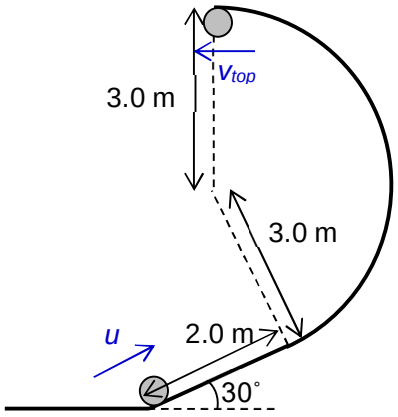
Figure 7.3 (not to scale)

(i) On Figure 7.4 below, sketch a graph to show how the magnitude of the centripetal acceleration a_c of the rollerblader varies with his linear speed v.



Figure 7.4 [1]

		L2	$a_c = \frac{v^2}{r}$ Since r is constant, the graph is a **positive quadratic (parabolic shape).**  	**B1**
		Examiner's comments: • A large number of candidates who attempted are able to provide correct answer.		
		(ii)	Determine the minimum speed *u* required for the rollerblader to reach the top of the circular ramp shown in Figure 7.3.	
			minimum speed = ……m s^{-1}	[4]

		L2	The minimum speed required occurs when the rollerblader *just* loses contact with the ramp at the top. Free body diagram of the rollerblader at the top:  W: weight of rollerblader N: normal contact force exerted by ramp on rollerblader **If he loses contact at this point, N = 0.**	**C1**
			W = m v_{top}^2 / r **mg = m v_{top}^2 / r** **v_{top} = √ (rg) = √ (3.0)(9.81)** = 5.4249 m s^{-1}	**M1**
			 	
			By conservation of energy, $\frac{1}{2}mu^2 = \frac{1}{2}mv_{top}^2 + mg[2.0sin30° + rcos30° + r]$ $\frac{1}{2}u^2 = \frac{1}{2}5.4249^2 + (9.81)[2.0sin30° + 3.0cos30° + 3.0]$ u = 12.605 = **13 m s^{-1} (or 12.6 m s^{-1})**	**M1** **A1**
		Examiner's comments: • A large number of candidates mistakenly did not consider the velocity of rollerblader at the top of the curved ramp. • A number of candidates just obtained the 2 marks for V_{top} as they also mistakenly took that as the initial minimum velocity.		
		(iii)	On Figure 7.5 below, sketch a graph to show how the kinetic energy of the rollerblader E_K varies with his vertical displacement above the base of the ramp s_y when he moves up the ramp with this minimum speed.	

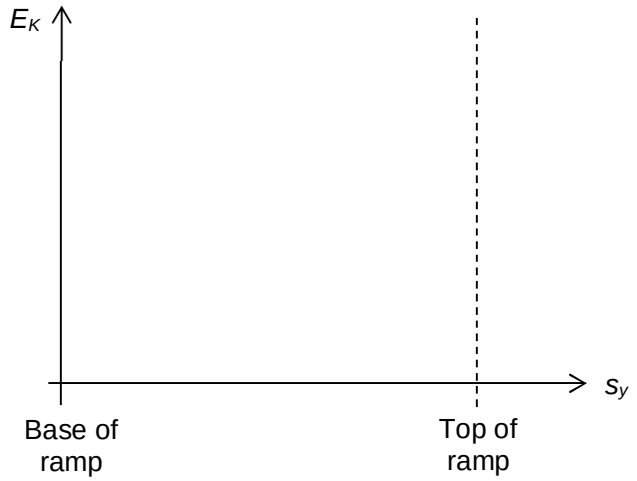


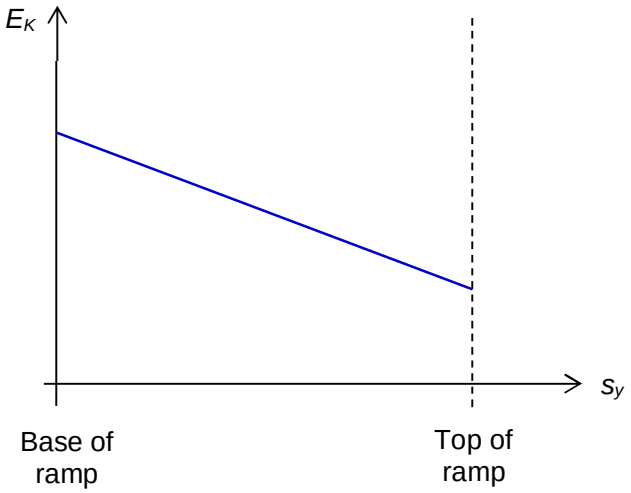
Figure 7.5 [2]

L3

GPE = mgs_y

$\Delta KE = -\Delta GPE$

Thus since ΔGPE varies linearly with s_y, ΔKE also varies linearly with s_y.



- **straight line, with negative gradient** **B1**
- **non-zero (since he moves up the ramp with the minimum speed, it means that his KE is always greater than zero)** **B1**

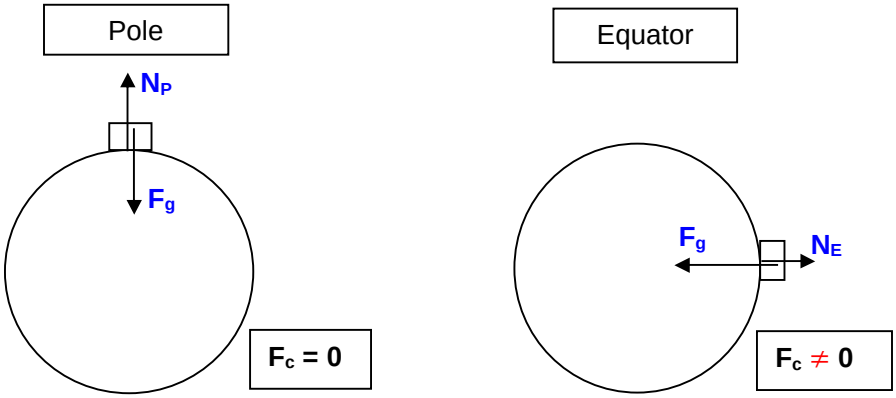
Straight line MUST be drawn using a ruler!

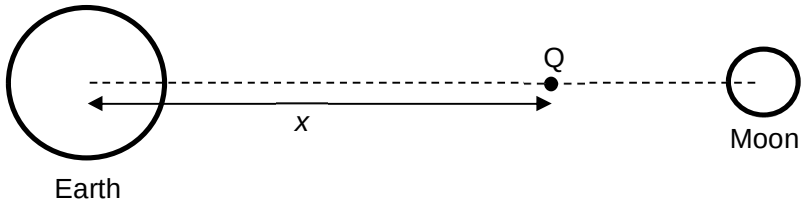
Examiner's comments:

- Poorly done. Few drew straight line or considered conservation of energy to obtain final answer.

[Total: 20]

8	**(a)**	**(i)**	The Moon is continuously attracted towards the Earth by gravitational force and orbits around the Earth. Explain why the Moon does not fall into the Earth.	
			……..…	
			……..…	
			……..…	
			……...	[2]
		L2	The **gravitational force exerted by Earth on Moon provides for the centripetal force required for Moon's orbital motion**, **which only changes the direction of the Moon's linear velocity and not its magnitude**. (The resultant force acting on moon is always perpendicular to the linear velocity of moon.)	**B1** **B1**
		Examiner's comments: • Poorly done • A lot of candidates have misconceptions on Newton's Second Law and Third Law and wrongly deduce that the net force on the moon is zero as both Moon and Earth experience the same magnitude of gravitational force. There is net force on the moon is the centripetal force which is provided by the gravitational force on it. • A number of candidates incorrectly discussed effect of gravitational force due to other masses (such as the Sun) on the moon which prevents the moon from falling into Earth. Considering the distance between moon and Earth versus the distance between our moon and Sun, this gravitational force on moon by the Sun will be much smaller (and its effect negligible). • A number of candidates also meandered into wrong logical reasoning by comparing the acceleration of the moon and Earth which would not explain why the moon is falling into the Earth.		
		(ii)	Assuming Earth to be a perfect sphere, explain why the weight of a person measured on an electronic balance at the poles is greater than that measured at the equator.	
			……..…	
			……..…	
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			……..…	
			……..…	

			…….....	[4]
		L3	Free body diagrams:  N: Normal contact force exerted by electronic scale on person F_g: Gravitational force exerted by Earth on person **Electronic balance reads the normal contact force N** exerted by the person on the electronic balance (i.e. the apparent weight of the person). **B1** **At a pole, person does not undergo circular motion, hence centripetal force F_c equals zero. Thus N equals in magnitude to F_g.** **B1** **At the equator, person undergoes circular motion, hence F_c is non-zero. Thus N is lesser in magnitude than F_g.** **B1** **Since the Earth is assumed to be a perfect sphere, its radius at the poles and equator are equal, hence F_g at both locations are assumed equal.** This implies that N at the poles is greater than N at the equator (and hence the apparent weight of the person is lesser at the equator). **B1**	
		Examiner's comments: • Poorly done. • Please take note **that symbols used in explanation have to be defined!** • Question stated to consider Earth as a perfect sphere, yet a number of candidates mistakenly take gravitational force to be not equal due to difference in radius of Earth at pole and equator. Hence very few obtained the last mark which requires student to state the actual gravitational force is a constant. • A number of candidates also did not show understanding that reading of the scale is the apparent weight, given by the magnitude of normal contact force. • A number of candidates gave the wrong expression of net force acting on the person and took normal contact force or gravitational force as the net force.		
	(b)	The Moon has an orbital period of about 27.3 days. The mass of the Earth is 6.00×10^{24} kg and the mass of the Moon is 7.34×10^{22} kg.		
		(i)	Show that the distance between the centres of the Moon and the Earth is approximately 383 000 km.	
				[3]

		L2	**Gravitational force F_G exerted by Earth on Moon provides for the centripetal force F_C required for Moon's circular motion.** $F_G = F_C$ $G\frac{M_E M_M}{r^2} = M_M r\omega^2$ $G\frac{M_E}{r^2} = r(\frac{2\pi}{T})^2$ $(6.67 \times 10^{-11})\frac{6.00 \times 10^{24}}{r^2} = r(\frac{2\pi}{27.3 \times 24 \times 60 \times 60})^2$ $r = 383.49 \times 10^6\ m$ $r = 383\,000\ km$ (3 s.f.)	**B1** **B1** **B1** A0
		Examiner's comments: • Since it is a show question, it is required that candidates show correct substitution of values to obtain final answer. • Incorrect substitution or incorrect equation used/incorrect equation manipulation will not be granted marks. • Please also explicitly state gravitational force by Earth on moon provides for centripetal force!! Symbols need to be defined!		
		(ii)	Hence, calculate the distance *x*, from the centre of the Earth along the Earth-Moon line to the point Q where the resultant gravitational field strength due to the Earth and Moon is zero.  **Figure 8.1** (not to scale)	
			x = ……m	[2]
		L1	As Q, $g_{net} = 0$, hence $\lvert g_E \rvert = \lvert g_M \rvert$ $G\frac{M_E}{x^2} = G\frac{M_M}{(383 \times 10^6 - x)^2}$ $\frac{M_E}{M_M} = \frac{x^2}{(383 \times 10^6 - x)^2}$ $\frac{6.00 \times 10^{24}}{7.34 \times 10^{22}} = \frac{x^2}{(383 \times 10^6 - x)^2}$ Solving, $x = 3.4486 \times 10^8 =$ **3.45×10^8 m**	**M1** **A1**

Examiner's comments:

- Mostly well done except for power of 10 error.
- Candidates who are not able to obtain the answer usually made the mistake of using wrong equation for g, or unable to do calculation to obtain final answer despite correct expressions and values used.

(iii) Calculate the gravitational potential at point Q.

gravitational potential at point Q = ……..…….J kg^{-1} [2]

L1 Gravitational potential at point Q, Φ_Q

$= -G\frac{M_E}{x} + \left(-G\frac{M_M}{(383\times10^6 - x)}\right)$

$= -(6.67 \times 10^{-11})\left[\frac{6.00 \times 10^{24}}{3.4486 \times 10^8} + \frac{7.34 \times 10^{22}}{(383 \times 10^6 - 3.4486 \times 10^8)}\right]$ **M1**

$= -1.2888 \times 10^6 = \mathbf{-1.29 \times 10^6\ J\ kg^{-1}}$ **A1**

Examiner's comments:

- Candidates who fail to obtain the mark here had used wrong formula or did not sum up the potential correctly (i.e. subtracted potential due to moon from potential due to Earth)
- Full ecf given if candidates showed correct substitution with correct expressions ad calculations based on their answer of x in b(ii)

(iv) The radius of the Earth is 6370 km and the radius of the Moon is 1740 km.

Determine the minimum energy required to send a stationary capsule of mass 30 kg from the surface of the Earth to the surface of the Moon.

			minimum energy = ……..……..J	[4]
		L2	Gravitational potential at the **surface** of the **Earth**, Φ_{ES} $= -\boldsymbol{G}\frac{\boldsymbol{M_E}}{\boldsymbol{R_E}} + \left(-\boldsymbol{G}\frac{\boldsymbol{M_M}}{(\boldsymbol{383\times10^6-R_E})}\right)$ $= -(6.67 \times 10^{-11})\left[\frac{6.00\times10^{24}}{6370\times10^3} + \frac{7.34\times10^{22}}{383\times10^6-6370\times10^3}\right]$ $= -6.2839 \times 10^7$ J kg^{-1} For **minimum** energy requirement, **capsule has to reach point Q, and has zero kinetic energy when it reaches point Q**. *[Recap: The point where g_{net} = 0 between Earth and Moon is also the point of maximum potential.]* **Change in** potential energy, $\Delta\Phi = \Phi_Q - \Phi_{ES}$ = **$-1.2888 \times 10^6 - (-6.2839 \times 10^7)$** Hence, the **minimum energy required** = m ΔΦ = **$(30)(-1.2888 \times 10^6 - (-6.2839 \times 10^7))$** = **$1.85 \times 10^9$ J** *Note: Enough energy must be supplied for the capsule to overcome the maximum increase in GPE in moving from Earth to Moon, which is from Earth's surface to the neutral point. Once the capsule passes the neutral point, it will be experience a net gravitational force towards the Moon which accelerates it towards the Moon without the need for external energy to be supplied.*	**M1** **M1** **M1** **A1**
		Examiner's comments: • Poorly done as majority of candidates did not consider the potential at surface of Earth due to the moon. • A number of candidates fail to see that capsule just has to reach neutral point Q, and did not consider potential at point Q to be the final position.		
	(c)	A planet of radius *R* is isolated in space. It has no atmosphere, and its mass can be assumed to be concentrated at its centre. A projectile is launched from the surface of the planet with a **minimum** kinetic energy ***E*** required for the projectile to **not** return to the planet. On Figure 8.2, draw:		

		(i)	a line, labelled **P**, to show how the gravitational potential energy of the projectile varies with distance from the centre of the planet as it moves from *R* to 4*R*,
		(ii)	a line, labelled **K**, to show how the kinetic energy of the projectile varies with distance from the centre of the planet as it moves from *R* to 4*R*.

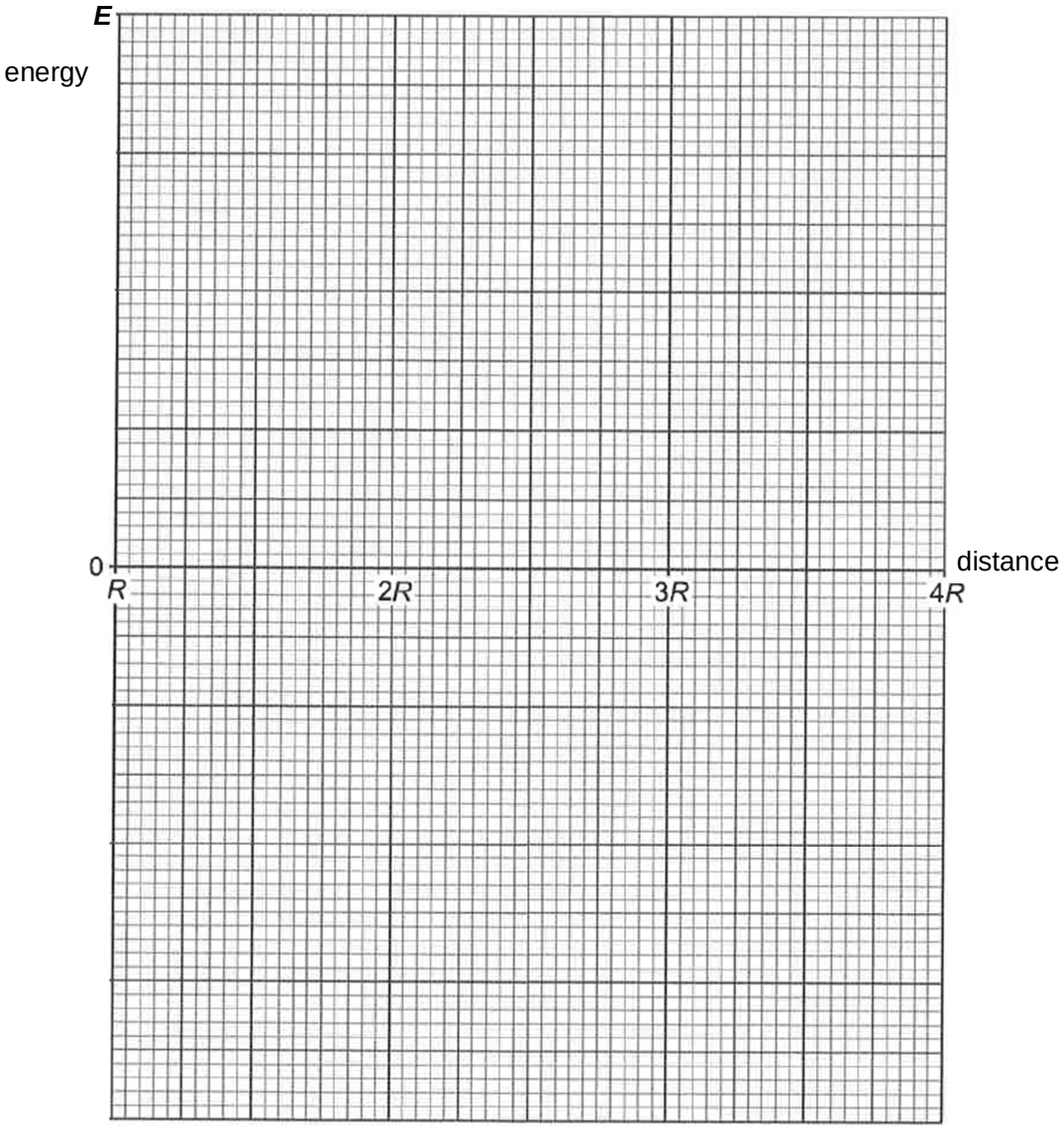


Figure 8.2 [3]

L2

Q: What is the initial value of GPE?

***Total energy is conserved**. Since $E_{T,\infty} = E_{K,\infty} + E_{P,\infty} = 0 + 0$, therefore **at every instant, $E_T = 0$**.*

*So, **$E_K + E_P = 0$ throughout**.*

*Thus, if $E_{K,\text{surface}} = E$, then **$E_{P,\text{surface}} = -E$.***

Q: What is the variation of GPE with distance from centre of planet like?

$E_P = -\frac{GMm}{r}$, *where GMm remains constant.*

$\therefore E_P \propto \frac{1}{r}$ *and negative. (i.e. when r is doubled, E_P is halved, etc.)*

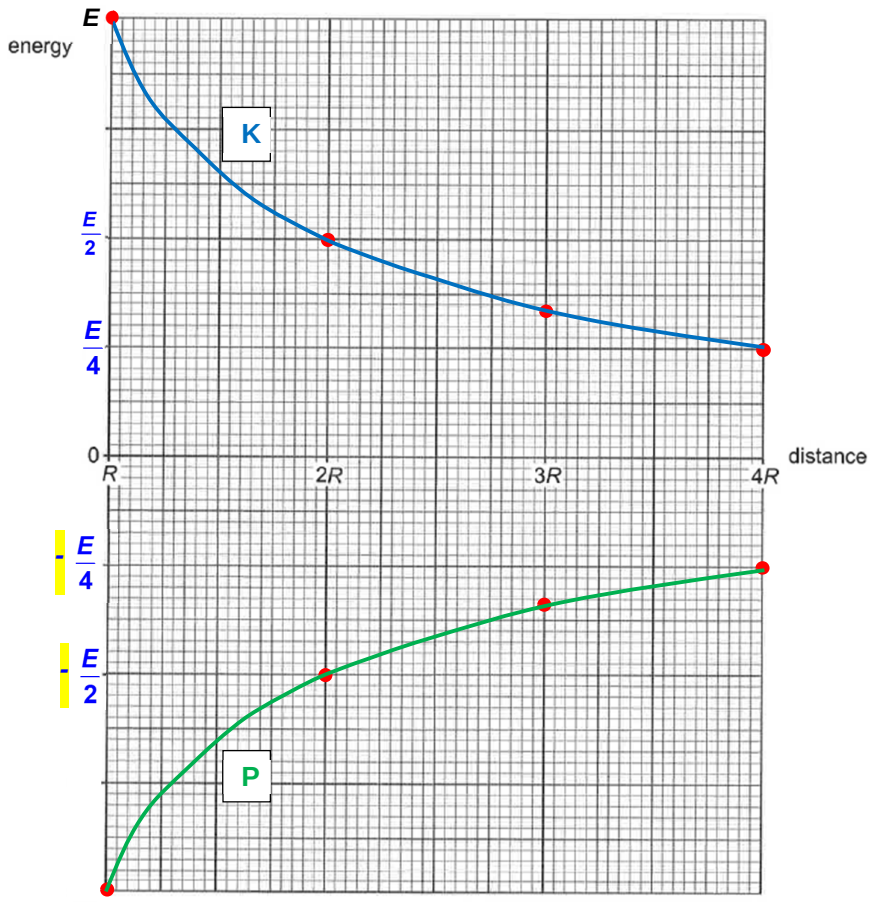
(i) *Tabulating,*

r	E_P	**Remarks**
R	$-E$	*Value of $E_{P,\,surface}$ found earlier.*
$2R$	$-\frac{E}{2}$	*20 small divisions vertically below horizontal axis*
$3R$	$-\frac{E}{3}$	*~13.3 small divisions vertically below horizontal axis*
$4R$	$-\frac{E}{4}$	*10 small divisions vertically below horizontal axis*

(ii) ***Since $E_T = 0$ throughout****, we can deduce the corresponding values of E_K at these instances. Tabulating,*

r	E_K
R	$+E$
$2R$	$+\frac{E}{2}$
$3R$	$+\frac{E}{3}$
$4R$	$+\frac{E}{4}$

E_P-r graph is a negative hyperbola; E_K-r graph is a positive hyperbola. And the sum of both graphs at every point equals zero since $E_T = 0$.



Marks scheme:

1 mark: Hyperbolic shape of both graphs, and, GPE negative while KE positive. Both graphs labelled as instructed. **B1**

		1 mark: Knowledge that TE = 0 throughout. ***1 mark: Precision of both graphs – GPE and KE values at R, 2R, 3R and 4R determined and plotted accurately to half-smallest division. [No penalty if y-axis not labelled, but points should be plotted accurately based on the E given.]***	**B1** **B1**
		Examiner's comments: • Poorly done. • A number of candidates would got 1 mark for shape of graph • For mark to be given for total energy = 0 , candidates need to show clearly that the shape of both graphs are reflection of each other about the horizontal axis of graph.	

[Total: 20]

END OF PAPER 3

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